part of a surveillance for rotavirus disease between September 2012 and April 2019 were tested for norovirus GI, norovirus GII, astrovirus, sapovirus and adenovirus 40/41 target by qPCR. The following regions/genes were targeted for each of the viruses-capsid region for astrovirus, ORF-1 for sapovirus, ORF1/ORF2 junction region for norovirus and hexon for adenovirus 40/41. Rotavirus was detected in these samples using Rotaclone antigen detection ELISA. All samples with positive amplification by qPCR assay for norovirus, astrovirus or sapovirus were further characterised and genotyped by sequencing pertinent regions of the virus genomes. Rotavirus vaccination with the Rotavac vaccine was introduced in the national immunisation schedule of India in a phased manner from 2016. For the state of Andhra Pradesh where Tirupati is located, rotavirus vaccination was introduced in April 2016.

Results: The median annual percentage of samples positive for rotavirus declined from 22.3% (range 17.02–34.12%) in the pre-vaccine introduction period to 14.36% (range 4.78–25%) in the post-vaccine period. For astrovirus and sapovirus, the median annual percentage positivity showed a significant increase in the post-rotavirus vaccine period (median annual positivity for astrovirus in post-vaccine period of 18.85% vs 7.89% in pre-vaccine period; median annual positivity for sapovirus in post-vaccine period of 17.33% vs 4.67% in pre-vaccine period). No such trends were seen for norovirus GI or norovirus GII. Adenovirus 40/41 was detected in a significantly higher proportion of cases in pre-vaccination era.

Conclusion: While a significant decline was seen in the proportion of acute gastroenteritis cases testing positive for rotavirus in the post-rotavirus vaccination years, other enteric viruses such as astrovirus and sapovirus were detected in a significantly higher proportion of cases in the post-vaccination era.

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Investigation of Hepatitis C virus among blood donors in Kwara state

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Background: Hepatitis C virus (HCV) is one of the implicated etiologies of Human hepatitis which is a major public health burden. A good proportion of the world's populations are carriers of HCV with a seemingly increasing number annually. This study screened for Hepatitis C Virus Ribonucleic Acid (HCV-RNA) from pre-screened blood at University of Ilorin Teaching Hospital (UITH), and determined the circulating genotypes using phylogenetic analysis of sequence results and possible risk factors.

Methods and materials: The study was carried out between April and August 2018 with a total of 144 participating blood donors. Socio-demographic variables were obtained using a pre-tested self-administered interviewer-based questionnaire. Whole blood samples were obtained, and an enzyme linked immunosorbent assay (ELISA) was carried out on the blood samples. HCV-RNA was extracted using RNA extraction kit and was amplified using reverse transcriptase polymerase chain reaction (RT-PCR) with specific primers. The amplified region of 405 base pair was purified and sequenced using Sanger sequencer, and a phylogenetic tree was constructed.

Results: The prevalence of HCV antibodies by ELISA was 6.9% (10/144) and the HCV-RNA was 18.1% (26/144), with genotypes 1 (21.4%), 2 (14.3%), 3 (21.4%) and 42.9% uncharacterized strain of HCV respectively from the genetic analysis of the samples. The relationship between the possible risk factors and HCV-RNA was not statistically significant except for the possible risk of transmission ($p = 0.023$).

Conclusion: It is evident that the use of ELISA alone for blood screening may not be reliable, and could be suggested that to reduce the burden of transfusion transmitted HCV, molecular technique should be embrace for screening of blood donors. The knowledge of the prevailing genotype will help tailor treatment options and vaccine development in the population. Further research should be carried out on other possible risk of infections and route of HCV transmission besides the known risk factors in Nigeria.

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Prognostic role of liver function tests in pediatric dengue illness

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Background: A series of biochemical changes occur during the course of dengue infection. Liver involvement in dengue infection is common and sometimes leads to acute liver failure (ALF), life threatening hemorrhage and fatal outcomes. Paucity of data on utility of liver function tests (LFT) in pediatric dengue illness. Therefore we aimed to determine liver dysfunction, kinetics of LFT and whether the extent of liver injury has any effect on the severity of dengue infection in children.

Methods and materials: A prospective study of hospitalized children (1–12 years) with dengue infection was conducted at Pediatrics ward of King George's Medical University Hospital, India. Daily blood samples were obtained from dengue confirmed cases from the time of admission till discharge for the assessment of liver function tests. Severity of hepatitis was graded into: mild (up to 5 times), moderate (5–10 times) and severe hepatitis (>10 times) based on alanine transaminase (ALT) elevation. The patients were classified into: non-severe dengue (NSD; dengue with or without warning sign) and severe dengue (SD). Events including organ dysfunctions and mortality were noted.

Results: A total of 102 children (boys, $n = 66$) median age 72 (12–144) months with confirmed dengue infection [non-severe dengue ($n = 41$) and severe dengue ($n = 61$)] were analyzed. Hepatomegaly and clinical jaundice was observed in 40% and 14% respectively. Elevated transaminases were most common biochemical abnormality seen in 92%. Aspartate transaminase (AST), ALT, alkaline phosphates (ALP) and gamma glutamyl transferase (GGT) levels were elevated in 87%, 82%, 22% and 26% respectively. Maximum abnormalities in LFT peaked around day 5 (AST, ALT, INR) and 7 (ALP, GGT) of illness. SD patients had significantly elevated transaminases in comparison to NSD (100% vs 80%; $p = 0.006$). Those with severe hepatitis had severe dengue illness and higher incidence of plasma leakage. Organ dysfunctions were commonly observed in severe hepatitis patients as compared to mild–moderate hepatitis in terms of altered sensorium ($p < 0.001$), ALF ($p < 0.001$), acute kidney injury ($p < 0.001$) and shock ($p < 0.001$). Fourteen patients died, 2/3rd from severe hepatitis ($p < 0.001$).